📄 Resume Department Mapping — End-to-End Process Documentation (Consulting Guide)

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1. Executive Summary

This project documents the full workflow of extracting, processing, and classifying student resumes into standardized departments. The objective was to create a **manual benchmark dataset** to validate an automated resume classification system.

The process includes:

- Resume extraction (Python + Power Query)
- Data structuring and cleaning
- Manual skill validation
- Rule-based department classification (Excel)

The final output is a **clean, explainable, and defensible dataset** aligned with business requirements.

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2. Objective

- Extract and structure resume data into a usable format
- Create a **manual benchmark** for automation validation
- Ensure classification accuracy using **structured logic**
- Deliver an **auditable and explainable workflow**

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3. End-to-End Workflow Overview

```
Raw Resume Files
        ↓
Python Extraction
        ↓
Power Query Transformation
        ↓
Excel Structured Dataset
```

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        ↓
Manual Skill Validation
        ↓
Department Mapping (Rule-Based)
        ↓
Final Benchmark Dataset
```

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4. Step 1 — Resume Extraction (Python)

Approach:

- Used Python to extract structured data from resumes
- Focused on:
  - Name
  - Degree
  - Skills
  - Relevant keywords

Purpose:

- Convert **unstructured resume text → structured dataset**

Output:

- Initial dataset of ~75 resumes

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5. Step 2 — Data Transformation (Power Query)

Actions Performed:

- Cleaned extracted fields
- Standardized text format
- Removed inconsistencies and noise
- Organized into tabular format

Purpose:

- Ensure data is **analysis-ready**
- Improve consistency before manual review

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6. Step 3 — Manual Benchmarking

Context:

Automation already performed classification.
Our task = **manually validate (benchmark)** the results.

Actions:

- Reviewed each resume individually
- Cross-checked:
  - Degree
  - Skills
- Ensured alignment with actual profile

Purpose:

- Create a **ground-truth dataset**
- Identify potential automation errors

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7. Step 4 — Skill Structuring

Actions:

- Consolidated skills into standardized format (Column E)
- Example:
  - “Python, SQL, Tableau”
  - “Finance, Research, Marketing”

Issue Identified:

- Some skills (e.g., “R”, “Excel”) appear across most resumes

👉 Led to classification bias in early stage

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8. Step 5 — Initial Classification Attempt (Issue)

Method:

- Skills-only classification using Excel formula

Result:

- Over-classification into:
  → INFORMATION SYSTEMS OPERATION MANAGEMENT

Root Cause:

- Common technical skills triggered same category repeatedly

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9. Step 6 — Key Improvement (Critical Fix)

Insight:

Skills alone are not reliable

Solution:

Implemented **Degree-First Classification Logic**

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10. Step 7 — Final Classification Model

10.1 Degree-Based Classification (Primary)

Degree Signal	Department
Management / Business Admin	MANAGEMENT DEPARTMENT
Accounting / Accountancy	ACCOUNTING & WRITING
Finance / Economics	FINANCE
Marketing / Communication	MARKETING
Analytics / Data / IS / CS	INFORMATION SYSTEMS OPERATION MANAGEMENT

Degree Signal	Department
Healthcare / Public Health	HEALTHCARE ADMINISTRATION

10.2 Skills-Based Classification (Fallback)

Skills Signal	Department
Accounting / Tax / Audit	ACCOUNTING & WRITING
Finance / Valuation / Modeling	FINANCE
Marketing / Brand / Research	MARKETING
Cybersecurity / AWS / Software	ITS
Python / SQL / Tableau / BI	INFORMATION SYSTEMS OPERATION MANAGEMENT

10.3 Default Rule

If no match:

→ MANAGEMENT DEPARTMENT

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11. Excel Implementation

Tools Used:

- `IF()`
- `SEARCH()`
- `ISNUMBER()`

Logic:

- Hierarchical evaluation
- Degree overrides skills
- Skills act as fallback

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12. Quality Control

Validation Performed:

- Spot-check multiple rows
- Verified:
  - Degree → correct department
  - Skills → correct fallback
- Ensured:
  - Balanced distribution
  - No over-classification
  - No missing values

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13. Key Design Decisions

Decision	Reason
Degree-first logic	Stronger signal than skills
Removed weak triggers (e.g., “R”)	Avoid bias
Structured fallback	Ensures completeness
Standardized outputs	Align with benchmark

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14. Limitations

- Keyword-based logic may miss nuanced profiles
- Some interdisciplinary cases may require manual override

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15. Outcome

- Created a **clean benchmark dataset**
- Improved classification accuracy significantly
- Built a **transparent and explainable system**
- Ready for comparison with automated model

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16. Strategic Insight (Consulting Angle)

This process demonstrates:

- Data pipeline design (Extraction → Transformation → Classification)

- Feature prioritization (Degree vs Skills)
- Bias identification and mitigation
- Rule-based modeling

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17. Next Steps

- Compare manual vs automated outputs
- Identify mismatches
- Refine classification rules
- Explore ML-based classification in future